signals such as voice, music, picture etc are transmitted is known as bandwidth. If v_1 and v_2 are the lower and upper-frequency limits of a signal, then the bandwidth, BW = $v_2 - v_1$.

v) Bandwidth of transmission system

The range of frequencies required to transmit a piece of specified information in a particular channel is called channel bandwidth or the bandwidth of the transmission system.

10.11 PROPAGATION OF ELECTROMAGNETIC WAVES

The electromagnetic wave transmitted by the transmitter travels in three different modes to reach the receiver according to its frequency range. They are

- Ground wave propagation (or) surface wave propagation (nearly 2 kHz to 2 MHz)
- Sky wave propagation (or) ionospheric propagation (nearly 3 MHz to 30 MHz)
- Space wave propagation (nearly 30 MHz to 400 GHz)

i) Ground wave propagation

If the electromagnetic waves transmitted by the transmitter glide over the surface of the Earth to reach the receiver, then the propagation is called ground wave propagation. The corresponding waves are called ground waves or surface waves. The pictorial representation is shown in Figure 10.53(a). Here both transmitting and receiving antennas must be close to the Earth's surface.

It is mainly used in local broadcasting, radio navigation, for ship-to-ship, ship-to-shore communication and mobile communication.

ii) Sky wave propagation

The mode of propagation in which the electromagnetic waves radiated from an antenna, directed upwards at large angles, gets reflected by the ionosphere back to Earth is called sky wave propagation or ionospheric propagation. The corresponding waves are called sky waves (Figure 10.53(b)).

Ionosphere acts as a reflecting surface. It is at a distance of approximately 50 km and spreads up to 400 km above the Earth's surface. Due to the absorption of ultraviolet rays, cosmic ray, and other high energy radiations like α, β rays from sun, the air molecules in the ionosphere get ionized. This produces charged ions and these

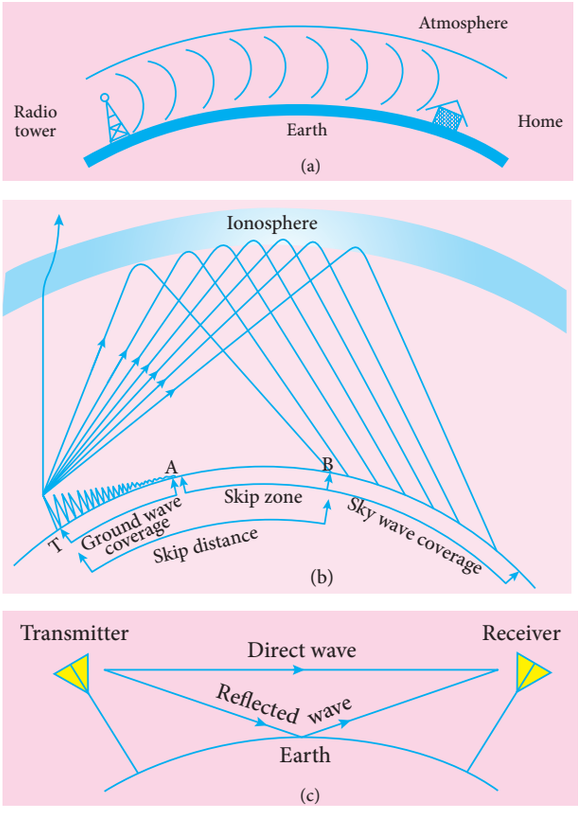


Figure 10.53 Propagation of EM waves (a) Ground wave (b) Sky wave (c) Space wave

XII_Physics_Vol-2_UNIT-10_Part 2_EM.indd 238 04-03-2022 17:19:20